

21CSC529T Inferential Statistics

Unit 1 : Introduction to Statistics

Syllabus :

Unit - 1: INTRODUCTION TO STATISTICS

Unit - 2: PROBABILITY

Unit - 3: ESTIMATION & HYPOTHESIS TESTING

Unit - 4: HYPOTHESIS TESTING -I

Unit - 5: HYPOTHESIS TESTING -II

Unit 1

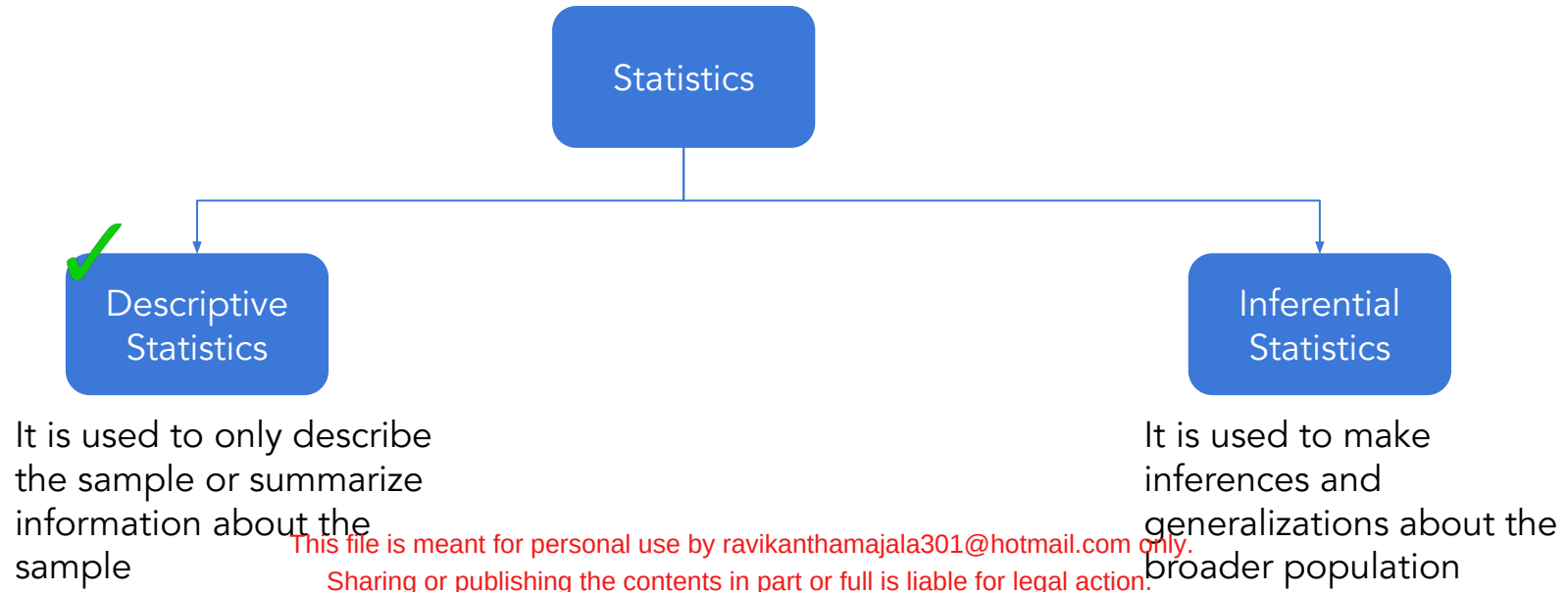
- Role of statistics in Data science
- Different types of data. - Understanding Random variable, Numerical Variable and Categorical Variable
- Data Collection, Types and its formats
- Types of Sampling
- Descriptive Statistics - Measure of Central Tendency Mean, Median and Mode
- Measure of Dispersion Range, Quartiles, Standard Deviation, Variance
- Distribution of Data Skewness and Kurtosis
- Covariance and Correlation, Difference between covariance and correlation and its significance.

Agenda

- Introduction to Statistics
- Classification of Data
- Various Types of Sources
- Sampling
- Types of Sampling
- Descriptive Statistics
 - Measure of Central Tendency
 - Measure of Dispersion

Statistics

Statistics is the branch of mathematics that deals with the collection, analysis, interpretation, presentation, and organization of data. It provides tools and methods for making sense of numerical data, allowing researchers and analysts to draw conclusions, make predictions, and inform decisions.



Statistics

Descriptive Statistics		Inferential Statistics	
Measures of Central Tendency	Measures of Dispersion	Hypothesis Testing	Regression Analysis
Mean	Range	Z test	Linear Regression
Median	Standard Deviation	F test	
Mode	Variance	T test	
	Absolute Deviation		

Characteristics of Statistics

1. Aggregation of facts
2. Affected by several causes
3. Numerically Expressed
4. Enumerate Estimate as accurate as possible
5. Collection by systematic manner
6. Collecting for planned purpose
7. Comparability

Functions of Statistics

1. Concrete Factual Presentations
2. Reduce complexity of raw data
3. Make comparison and draw conclusions
4. Helps in forming Hypothesis
5. Making Predictions
6. Useful for Decision Making

Limitations of Statistics

1. Limited to quantitative study
2. Aggregate Measurements
3. Homogeneity of data
4. Results are true only on Estimates and not accurate numbers
5. Leads to erroneous conclusions

Descriptive Statistics

The Data

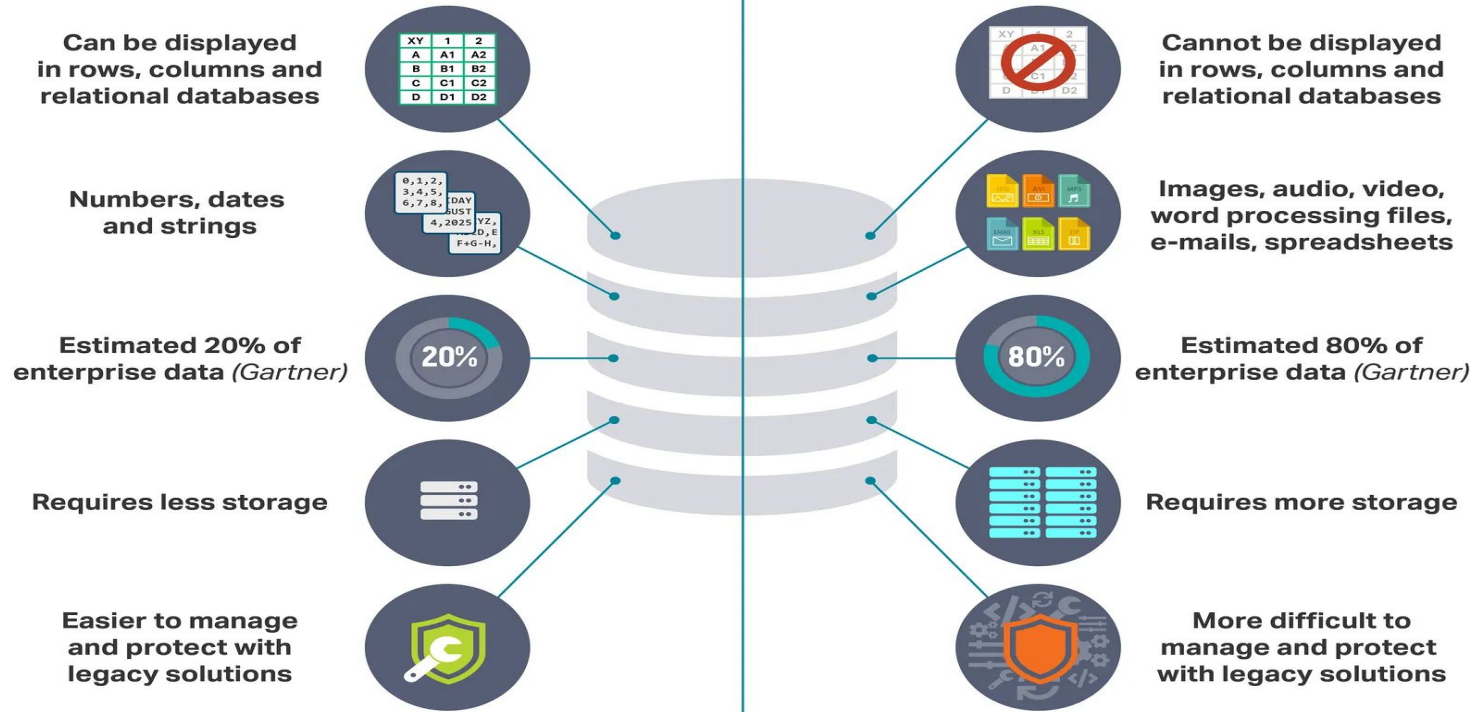
What is Data?

Data is units of information in structured or unstructured format.

Examples:

- Collection of relevant tweets
- Records of yield in a farm over a period of time
- Records of stock price every minute
- Records of performance of a sports person

Structured Data vs Unstructured Data



Sources of Data Collection

Primary Sources

- Data collected by user
- Raw data gathered first-hand from the source
- Objective for data collection is pre-defined

Secondary Sources

- Data is not collected by the user
- Data collected by someone else and is readily available
- Objective for the data collection could be different than the objective of the user

Various forms of Data

1. Textual Data
2. Numerical Data
3. Date time Data
4. Audio Data
5. Video Data
6. Log / Data File
7. Tabular Data
8. Survey / Questionnaire Data
9. Geospatial Data

Classification of Data

In general, there are four types of data. They are:

- Time Series Data
- Cross sectional data
- Pooled Data
- Panel Data

Time series data

- Time series data is the set of observations on a variable at different time points
- The data may be collected daily, weekly, monthly, or annually

Date	Prev.Close	Open.Price	High.Price	Low.Price
01-Jan-1996	408	407	407.9	405
02-Jan-1996	407.9	407	409	406.25
03-Jan-1996	406.25	409	409	409
04-Jan-1996	409	405	407	405
05-Jan-1996	406.3	401.5	401.5	401.5
08-Jan-1996	401.5	401.5	405	402
09-Jan-1996	404	404	400	395
10-Jan-1996	399.5	399.5	397	395
11-Jan-1996	396	399	405	396
12-Jan-1996	405	403	403	400

Sample data of daily
stock price of a
company



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Cross-sectional data

Cross-sectional data are set of observations on two or more variables at the same time point

Produce in the 2019 in (quintals)			
Region	Rice	Wheat	Maize
Region 1	28.76	86.66	8.23
Region 2	84.03	65.45	41.94
Region 3	71.75	28.99	38.43
Region 4	14	68.87	7.04
Region 5	43.97	54.27	86.51

Sample data of a farm produce in five regions in the year 2019

Pooled data

Pooled data is combination of both time series data and cross sectional data.

	Produce in the 2011		Produce in the 2012	
Appliance	Production	Profit (in '000 Rs)	Production	Profit (in '000 Rs)
Radio	280	13	121	8
Television	840	645	1456	5192
Washing Machine	775	2899	152	5706
Computer	876	6887	1005	6349
Air Conditioner	439	5427	328	8095

← Sample data of the number of appliances produced in an industry for two years

Panel data

- Panel data is type of pooled data
- The same cross-sectional unit is surveyed over time
- Also known as longitudinal or micro-panel data

Appliance	Year	Production	Profit (in '000 Rs)
Radio	2010	280	13
Television	2010	840	645
Washing Machine	2010	775	2899
Computer	2010	876	6887
Air Conditioner	2010	439	5427
Radio	2011	234	123
Television	2011	835	645
Washing Machine	2011	564	2899
Computer	2011	874	6887
Air Conditioner	2011	435	5427

Sample data of the number of appliances produced and profits earned by a company over a period of two years

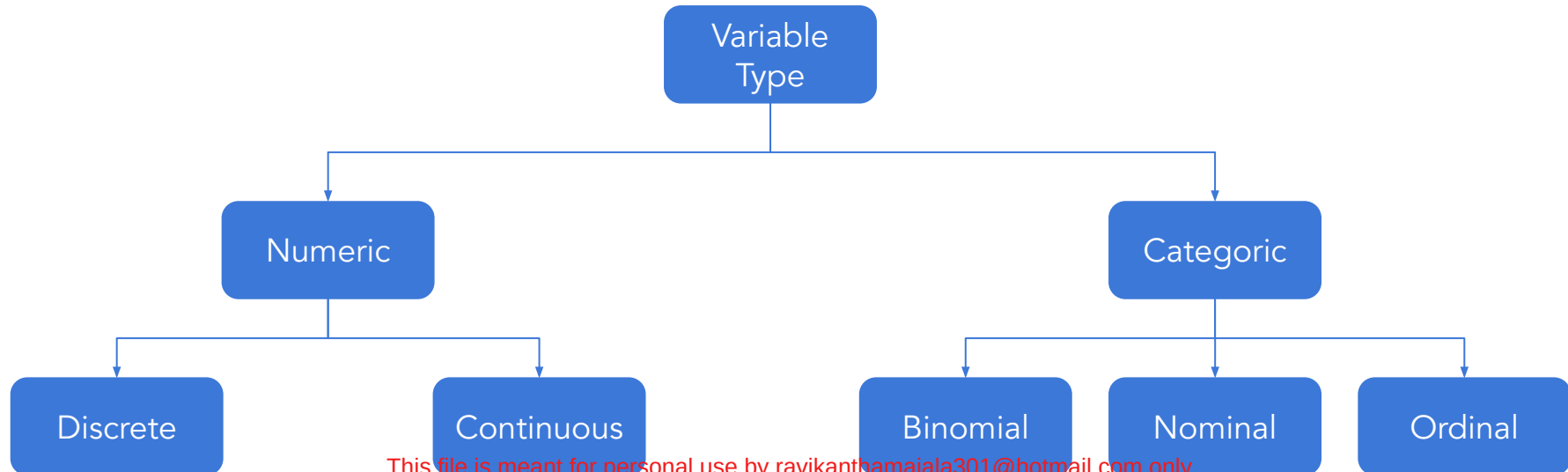
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Understanding the variable

Understand the variable

To understand the descriptive statistics, we need to first understand the type of each variable or column



Numerical Variable

A **continuous numeric variable** is a variable which can have infinite number of values within a range

Examples:

1. The amount of rainfall in millimeters
2. Price of a stock

Numerical Variable

- A variable which takes a numeric value is called **a numeric variable**
- Also known as **quantitative variable**
- A **discrete numeric variable** is a random variable which takes discrete values, i.e. values from the set of whole numbers only. It can take countably finite values

Examples:

1. Number of cars passing by a toll-gate every one minute
2. Number of defective items in a box

Categorical Variable

- Categorical variable has two or more levels
- Also known as **qualitative variable**

Examples:

1. Colors: red, orange, yellow
2. Gender: male, female
3. Economic status: low, medium, high

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Categorical Variable

Nominal Data:

- Nominal data has **no order** and has **two or more than two categories**
- It represents discrete units and are used to label variables
- Example: Housing type - Apartment, Bungalow, Penthouse

Binary Data:

- Binary data has **no order** and has **strictly two categories**
- Also known as dichotomous variable
- Example: Presence or absence of a disease

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Categorical Variable

Ordinal Data:

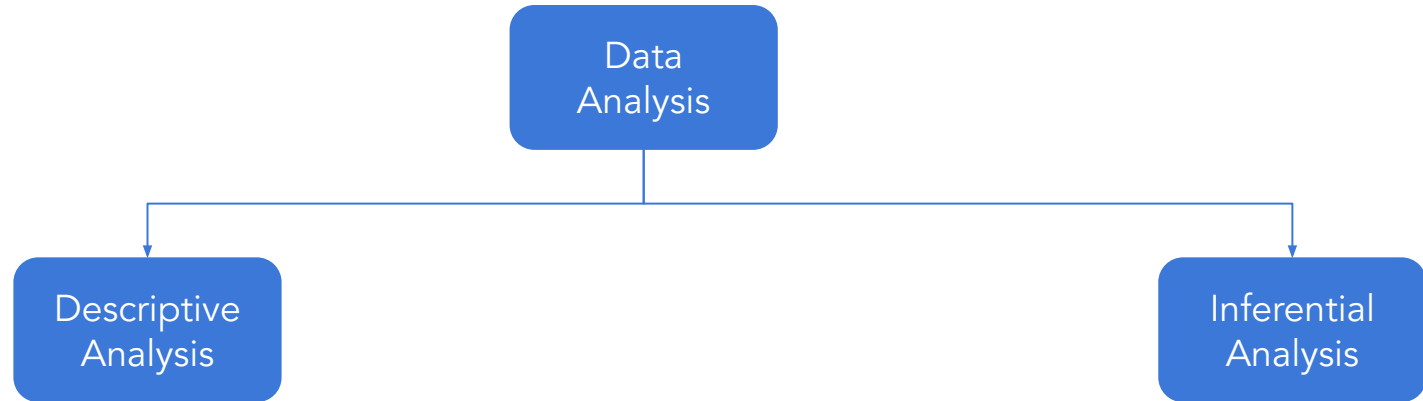
- Ordinal data is **ordered** nominal data
- It represents discrete and ordered units
- Example: Education - Primary, Secondary, High School, College and University which are ordered



- At times data may appear to be numeric but is actually categorical
- Consider the adjoining table. The 'Gender' column has value 1 and 0
- The numeric values used to represent a categorical variable cannot be used for numerical computation

Name	Gender	Age
Ria	1	45
Arya	1	23
Sam	0	34
Joe	0	54
Harry	0	12
Sarah	1	43

Data Analysis



- It is used to only describe characteristics of population using sample or summarize information based on sample

- Also known as Descriptive Statistics

- It is an analytical tool used to make inferences and generalizations about the broader population

- Also known as Inferential Statistics

Descriptive Statistics

Exploratory Data Analysis (EDA)

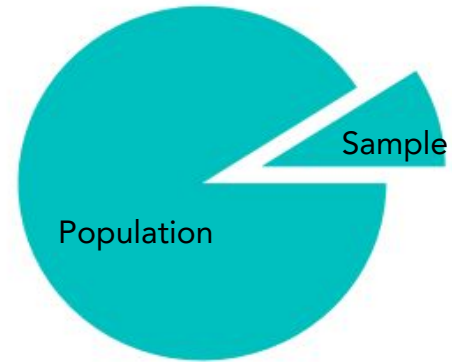
- EDA is a process of analyzing the datasets to summarize their main features using numerical and visual methods
- The numerical methods includes descriptive statistics and inferential statistics

What is descriptive statistics?

- Descriptive statistics is the term given to the analysis of the data that helps describe, visualize or summarize the data
- Helps exhibit the patterns in the data
- However, descriptive statistics only help in understanding the data and not make conclusions regarding any hypothesis
- We can not generalise the facts, they are confined only to the data at hand (that may be the **sample**)

Population & Sample

- **Population** is a collection of all the individuals or objects
- **Sample** is a subset of the population which is a representative of the population



Example:

Suppose a company producing electric bulbs wants to know the average life of a bulb. If the details on average life of all bulbs are available, then this information is regarded as the population.

It would be challenging for the company to test each and every bulb produced. In such a scenario, the company would draw a sample from the produced bulbs to test.

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Sampling

- Sampling is the process of selecting a subset of individuals, items, or observations from a larger population to make inferences or predictions about the entire population.
- The goal of sampling is to gather data in a way that accurately reflects the population as a whole
- Sampling save time, effort, and resources compared to studying the entire population.

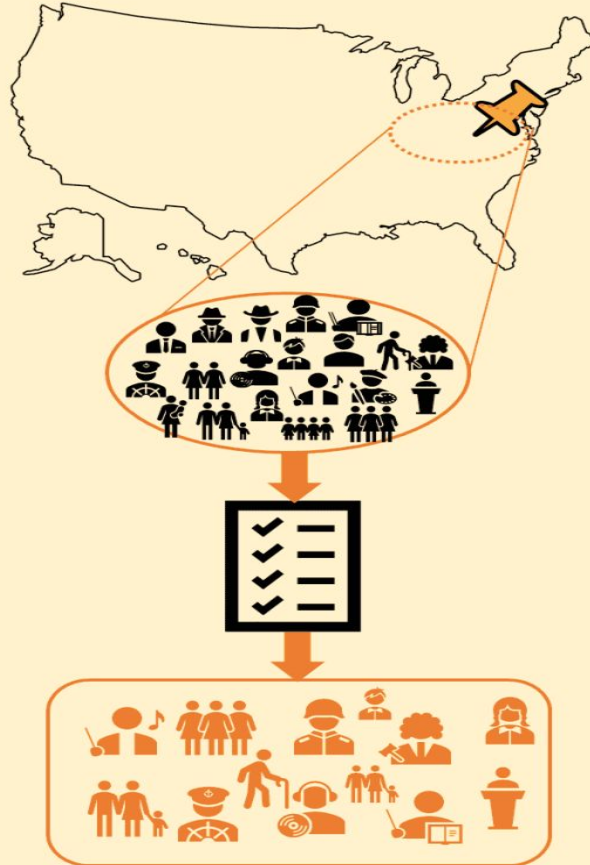
SAMPLING

Target Population

Study Population

Sampling frame

Sample



Generalization
of results

Inference

Reliable and
Valid data

Sample
statistic

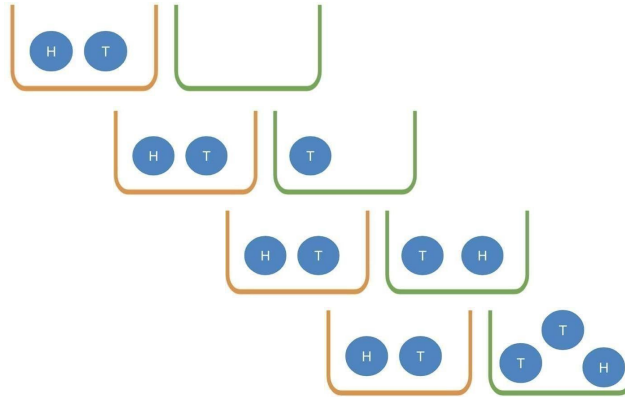
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Significance of Sampling

1. Resource Optimisation
2. Improved Data Accuracy
3. Facilitates Generalization
4. Ethical Considerations
5. Enhances Feasibility

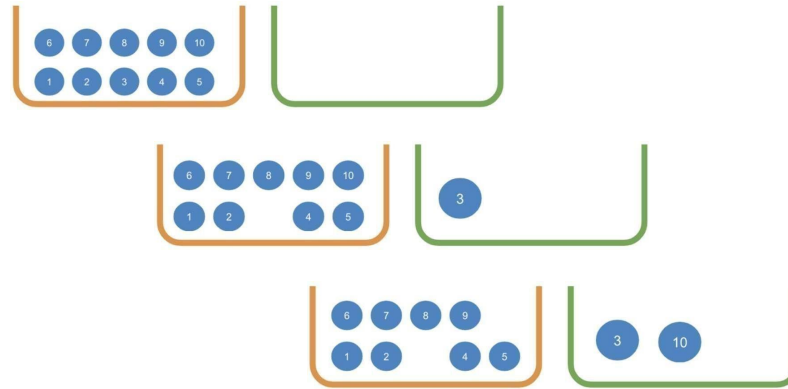
Sample with replacement

The unit selected to be in the **sample** is returned to the **population**.

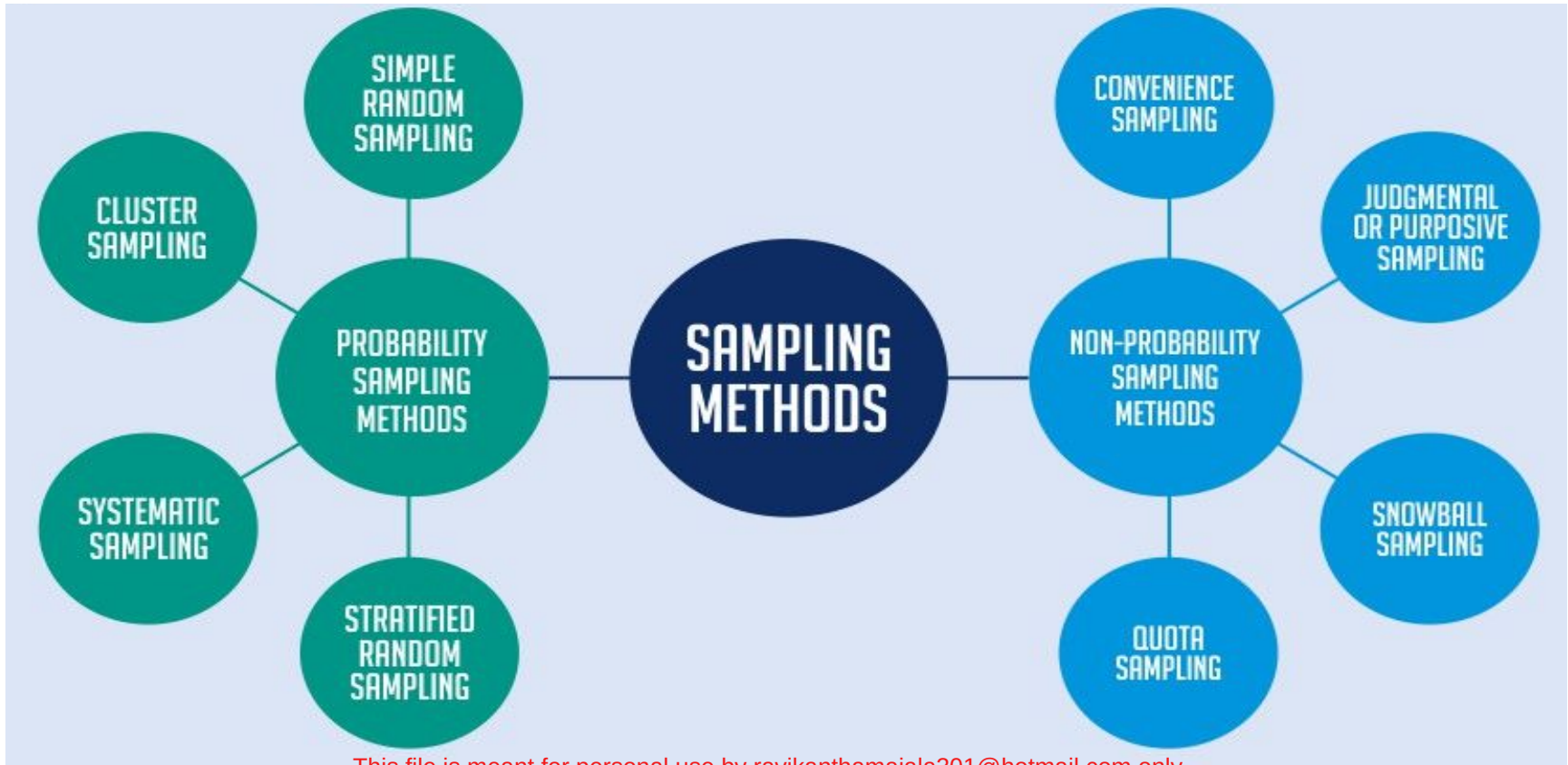


Sample without replacement

The unit selected to be in the **sample** is not returned to the **population**.



Types of Sampling:



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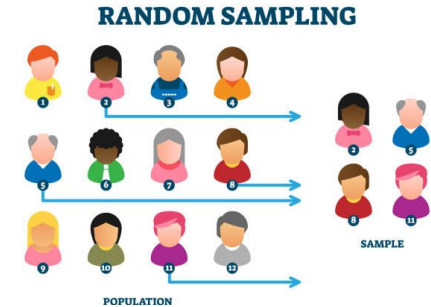
Probability Sampling

- Probability sampling is a sampling technique where a researcher selects a few criteria and chooses members of a population randomly.
- All the members have an equal opportunity to participate in the sample with this selection parameter.

Types of Probability Sampling

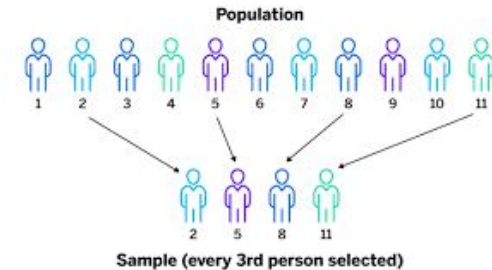
Simple Random Sampling:

- Every member of the population has an equal chance of being selected.
- Respects the principle of independence
- Example: random sample of 5 students from a class of 50 students



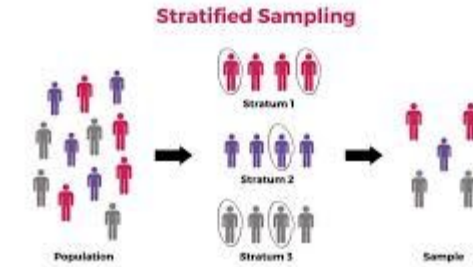
Systematic Sampling:

- Selects every k^{th} individual from a list or sequence.
- Example: Surveying every 3rd person entering a store.



Stratified Sampling:

- The population is divided into strata (groups) based on a characteristic (e.g., age, gender), and a random sample is taken from each stratum.
- In Stratified sampling we assume strata is homogenous.
- Example: Ensuring representation of different age groups in a survey.



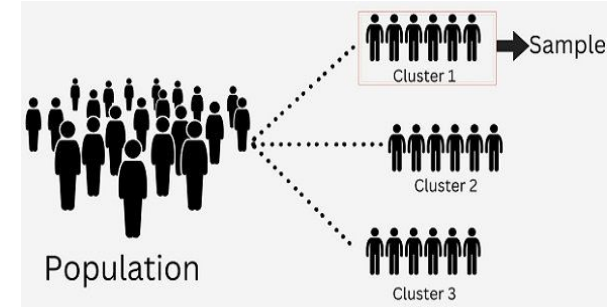
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Types of Probability Sampling

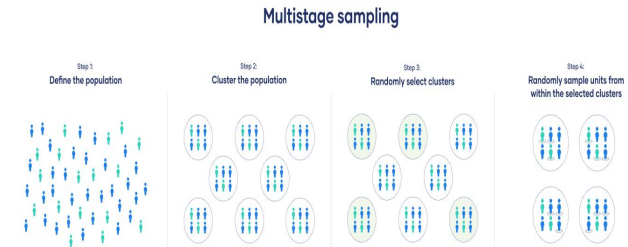
Cluster Sampling:

- The population is divided into clusters (e.g., geographical areas), and a random sample of clusters is selected, with all individuals within those clusters surveyed.
- Naturally formed group with diverse samples within group.
- Example: Selecting certain schools and surveying all students within those schools.



Multistage Sampling:

- Combines several sampling methods, typically involving selecting samples in stages, e.g., selecting clusters first, then using simple random sampling within each cluster.
- Example: First selecting cities, then schools within those cities, and then students within those schools.



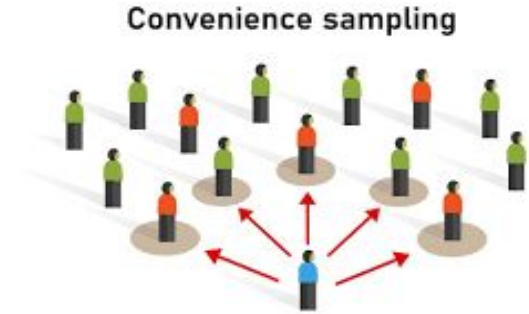
Non-Probability Sampling:

- In non-probability sampling, not every member of the population has a chance of being included, and the selection is often based on convenience or judgment rather than randomness.
- This method is often used when probability sampling is impractical.

Types of Non-Probability Sampling:

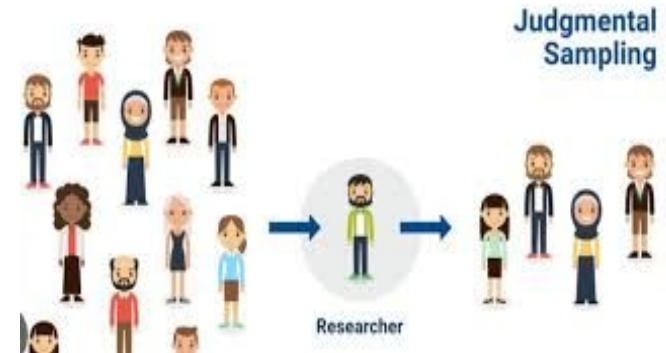
Convenience Sampling:

- The sample is taken from a group that is easy to access.
- Example: Surveying people at a mall because it's convenient.



Judgmental or Purposive Sampling:

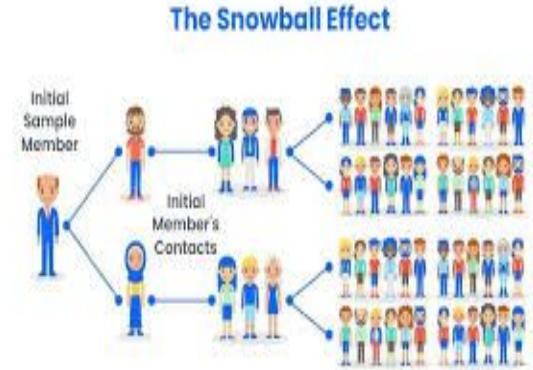
- The researcher selects the sample based on their judgment of what is most representative or typical.
- Example: Choosing experts in a field for a study.



Types of Non-Probability Sampling:

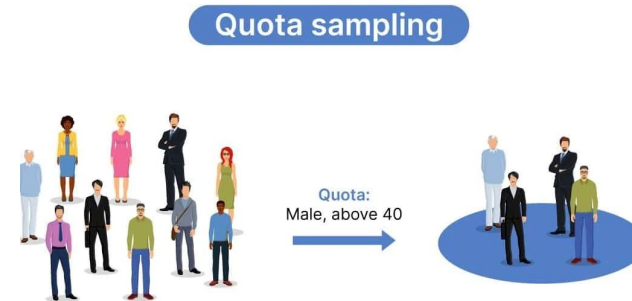
Snowball Sampling:

- Existing study subjects recruit future subjects from among their acquaintances.
- Example: A study on a rare disease where current participants refer others with the same condition.



Quota Sampling:

- The researcher ensures that the sample reflects certain characteristics of the population, but selection within those quotas is non-random.
- Example: Ensuring a sample includes a specific percentage of males and females, but selecting individuals conveniently.



Types of Sampling Error

- Random error

Random errors are unsystematic, unpredictable, and just as likely to over-as under-estimate the “true” score.

Difference between sample mean and true mean

Can be reduced by selecting large sample size

- Non - Random error

Non-random errors are systematic in over or under-estimating the true score

It is error in sampling process while collecting data

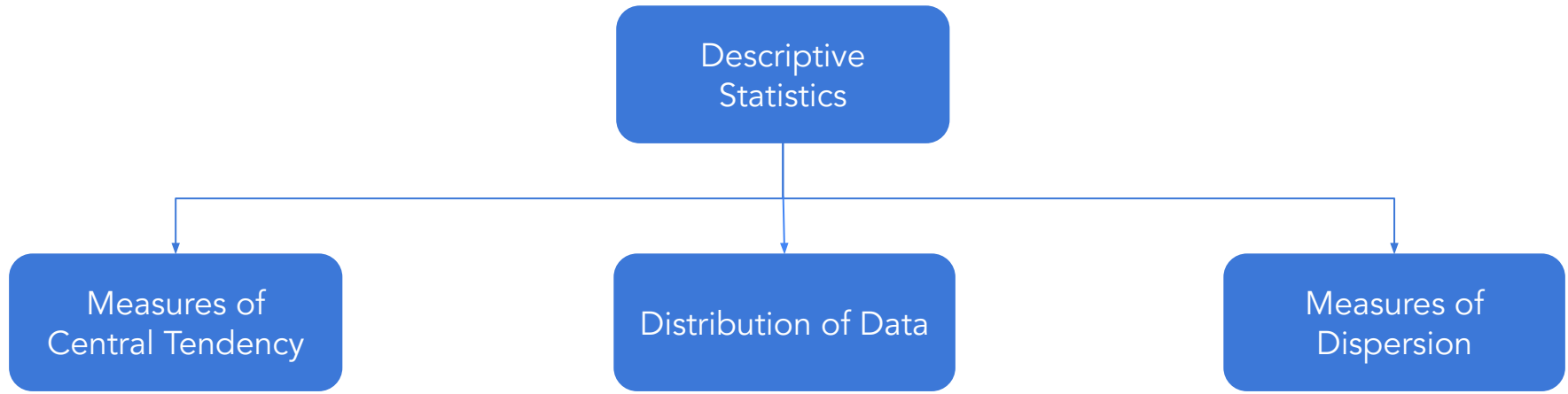
1. Sample bias
2. Completion rate
3. Interviewer effect

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Descriptive Statistics

Descriptive statistics is the term given to the analysis of the data that helps describe, visualize or summarize the data



Descriptive statistics

The major characteristics we observe in each variable of the data set are:

1. Measures of central tendency
2. Measures of dispersion

The above characteristics help us in understanding the data and are part of the exploratory data analysis

Measures of Central Tendency

Measures of central tendency

- A measure of central tendency is a single value that identifies the central position of the data
- It includes mean, median, mode and partition values

Mean

- Mean is a central tendency of the data
- It is defined as the sum of all observations divided by the number of observations:

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{N}$$

where N is the total number of observations

- The whole data is spread out around the mean

Mean

Merits:

- Based on all observation

Demerits:

- Affected by extreme observations (outliers)
- Arithmetically it is not possible to obtain the mean of a data with missing values
- Mean can not be calculated for categorical data

Obtain the mean

Each column of a DataFrame is a pandas Series.

Let us create a series and find its mean.

```
prices = pd.Series([0,0,35,40,10,54,87,12,95,64,56,4,45,76,34,56,  
87,34,56,32,56,48,89,42,65,100,99,98,100,96,99])
```

```
# obtain the mean of 'prices'  
prices.mean()
```

```
57.064516129032256
```

Obtain the mean

Arithmetically, it is not possible to compute mean of missing data. But, python ignores the missing values and calculates the mean of the available data in the series

Introduce missing values in
the previous series

```
# let us introduce some missing values in the series
# the remaining values are same as the series 'prices'
prices_missing = pd.Series([0,0,35,40,10,54,87,np.NaN,12,95,64,56,4,45,76,34,np.NaN,56,
                           87,34,56,32,56,48,np.NaN,89,42,65,100,99,98,100,96,99])
```

```
# obtain the mean of prices_missing
prices_missing.mean()
```

```
57.064516129032256
```

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Mean of series with missing values

We notice that the means obtained are the same

This is because the function drops the missing values before calculating the mean value

```
# obtain the mean of 'prices'  
prices.mean()
```

```
57.064516129032256
```

```
# obtain the mean of price_missing  
prices_missing.mean()
```

```
57.064516129032256
```


Trimmed mean

- Arithmetic mean obtained by ignoring lowest and highest $\alpha\%$ observations is called as the $\alpha\%$ - trimmed mean
- Using a trimmed mean, it is possible to eliminate the influence of outliers
- Note the observations are arranged either in increasing or decreasing order
- Also called as **truncated mean**

Example: The trimmed means are used in scoring the performance of an athlete during olympic games, to minimize the extreme scoring from possible biased judgements.

Obtain the trimmed mean

While working in python, we can directly use the built in function, `trim_mean()`, available in the library `scipy`.

```
import scipy
from scipy import stats

# obtain a 20% trimmed mean
scipy.stats.trim_mean(prices, proportiontocut =0.20)

58.89473684210526
```

Import the required library

Parameter 'proportion to cut' denotes the $\alpha\%$ values to exclude. It takes value between 0 to 0.51

Geometric Mean

Geometric mean is the n^{th} root of the product of n values of a set of observations.

$$\text{G.M} = (x_1 * x_2 * x_3 * \dots * x_i)^{(1/n)}$$

It is more appropriate than A.M in describing growth rates in finance, business etc. For eg: In business G.M is used to predict Compound annual growth rate (CAGR)

Geometric Mean

Suppose you invested in a stock, and the annual returns for the last three years were as follows:

Year 1: 10% (or 1.10 as a multiplier)

Year 2: 50% (or 1.50 as a multiplier)

Year 3: -20% (or 0.80 as a multiplier)

Soln:

$$\text{A.M: } (10+50-20)/3 = 13.33$$

$$\text{G.M: } (1.10 \times 1.15 \times 0.8)^{(1/3)} = 1.32^{(1/3)}$$

$$\text{G.M.R} = (1.098 - 1) \times 100 = 0.098 \times 100 = 9.8 \% \text{ Return}$$

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What is Geometric Mean Return for 20%, -10% and 30% return of X organisations.

$$\begin{aligned} \text{G.M.R} &= ([(1 + 0.2) * (1 - 0.1) * (1 + 0.3)]^{(1/3)} - 1) \\ &\approx (1.1197 - 1) = 0.119 * 100 = 11.9\% \end{aligned}$$

Harmonic Mean

Harmonic mean is the inverse of the arithmetic mean of the reciprocals of the observations of a set.

$$H.M = n / ((1/x_1) + (1/x_2) + (1/x_3) + \dots + (1/x_i))$$

Harmonic Mean

Find the average speed for a car that travels two equal distances at different speeds. The car travels the first half of the distance at 60 mph and the second half at 40 mph.

$$H.M = 2 / ((1/60) + (1/40)) \approx 48 \text{ mph.}$$



General Averages: Calculating average test scores, heights, weights, salaries, etc.
Calculating average income without the influence of extreme values (very high or low incomes).
Financial Ratios: Analyzing financial ratios like price-to-earnings (P/E) ratio.
Quality Control: Analyzing data with outliers, such as manufacturing defects.
Investment Returns: Calculating compound annual growth rate (CAGR) for investments.
Index Numbers: Constructing index numbers like Consumer Price Index (CPI).
Rates and Speeds: Calculating average speed for a journey with different speeds for different parts.
Finding average points, rebounds, or batting averages.

A.M
T.M
H.M
T.M
G.M
G.M
H.M
A.M

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Median

- Median is the middle most observation in the data when it is arranged either in ascending or descending order of their values
- It divides the data into two equal parts. Thus, it is a **positional average**
- Median will be the middle term, if the number of observations is odd
- Median will be average of middle two terms, if number of observations is even



Median

Odd Number of Samples

Example Data Set: 5, 3, 8, 1, 7

1. **Sort the Data:** 1, 3, 5, 7, 8
2. **Find the Median:** Since there are 5 numbers (an odd number), the median is the middle number, which is 5.

Even Number of Samples

Example Data Set: 12, 7, 9, 5

1. **Sort the Data:** 5, 7, 9, 12
2. **Find the Median:** Since there are 4 numbers (an even number), the median is the average of the two middle numbers. So, the median is $(7 + 9) / 2 = 8$.

Median

Merits:

- It is not affected by extreme values

Demerits:

- It is not based on all observations

Obtain the median

```
prices.median()
```

```
56.0
```

Interpretation: The median value is 56. It implies that the value 56 divides the data in two equal parts. There are 50% of the observations above and below this price.

Using mean and median

- For a numeric variable, the missing value is replaced by the mean if there are no outliers present
- In case if outliers are present the missing values are replaced by the median
- For symmetric data median = mean, so either of the value can be used

Mode

- Mode of the data is the value which has the highest frequency
- In other words, it is most repeated observation
- A set of observations having two modes is said to be **bimodal**
- A set of observations may have more than two modes. Such data is said to be **multimodal**

Mode

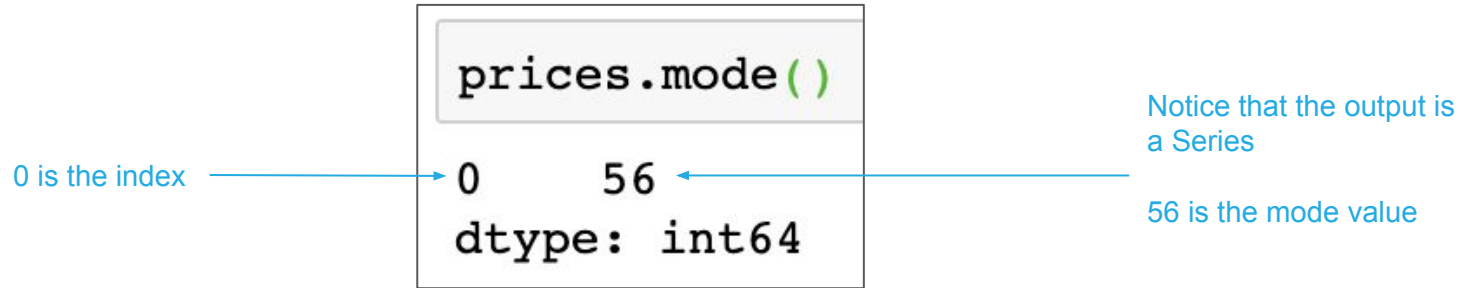
Merits:

- It is applicable to both numeric and categorical data
- It is not affected by extreme observations

Demerits:

- It is not based on all observations

Obtain the mode



Interpretation: The mode value is 56. It implies that the value 56 occurs the most number of times

Also note that there is only one mode. The data is **unimodal**

Obtain the mode

```
# create a series of pen prices
prices_pens = pd.Series([0,35,40,10,54,35,100,100])

# display the value counts of the
prices_pens.value_counts()
```

100	2
35	2
54	1
40	1
10	1
0	1

dtype: int64

Number of times each value is appearing in the data

Notice values 35 and 100 each occur 2 times

```
# print the mode
prices_pens.mode()
```

0	35
1	100

dtype: int64

The mode values

Interpretation: The mode values are 35 and 100. It implies that these values occur equal number of times and have the highest frequency.

Since there are two modes, the data is **bimodal**.

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Obtain the mode

```
# create a series of pen prices
prices_pens = pd.Series([0,0,35,40,10,54,76,34,56,87,34,56,100,99,100,99])

# display the value counts of the
prices_pens.value_counts()
```

```
56      2
99      2
100     2
34      2
0       2
76      1
10      1
40      1
87      1
54      1
35      1
dtype: int64
```

Number of times each value is
appearing in the data

Notice values 0, 34, 56, 99, 100
each occur 2 times

```
# print the mode
prices_pens.mode()
```

```
0      0
1     34
2     56
3     87
4     99
5    100
dtype: int64
```

The mode values

Interpretation: The mode values are 0, 34, 56, 87, 99, 100. It implies that these values occur equal number of times and have the highest frequency.

Also note that there are 6 mode values. Data is **multimodal**.

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Summary

- Mean, median and mode can be used to impute the missing values in the data
- For a symmetric distribution, mean = median = mode
- For any data,

$$Q2 = D5 = \text{Median}$$

- Mean is affected by the presence of outliers
- Trimmed mean is useful when the outliers are present in the data
- Median and mode are not affected by the presence of outliers

Thank you!

Happy Learning :)

Agenda

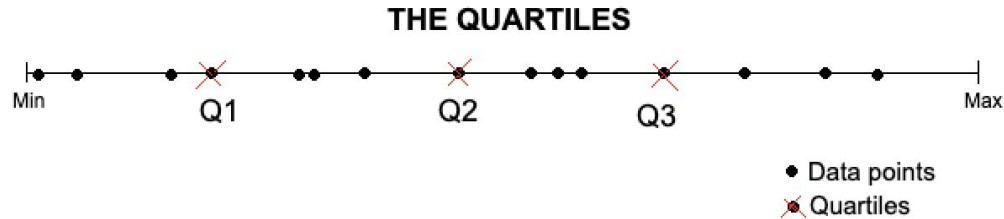
- Descriptive Statistics
 - Measure of Central Tendency ✓
 - Measure of Dispersion
 - Measure of Distribution - Skewness and Kurtosis
 - Covariance and Correlation

Partition values

- Partition values are the values which divide the dataset into equal parts
- Note that partition values may not be equispaced
- Also known as **quantiles**
- We have seen that the **median** divides the data into two equal halves. Similarly in order to make four parts we use **quartiles**, for ten parts we use **deciles**

Quartiles

- The values that divide the dataset into four equal parts are called quartiles



- 25% of the data lies below the first quartile (Q1) and 75% above it
- The second quartile divides the data points in two equal halves. Q2 is the median
- The third quartile (Q3) divides the the observation in 75%-25% meaning that 75% of the data lies below Q3 and 25% above it



Consider the dataset: [7, 9, 15, 18, 21, 24, 27]. Find Q1, Q2, Q3

$$Q1 = \frac{1}{4}(n+1)$$

$$Q2 = \frac{1}{2}(n+1)$$

$$Q3 = \frac{3}{4}(n+1)$$

Summary :

Q1: 9 (25% of the data is less than or equal to 9)

Q2: 18 (50% of the data is less than or equal to 18, which is also the median)

Q3: 24 (75% of the data is less than or equal to 24)

Obtaining the quartiles

The partition values are obtained using the quantile function specifying the percentage of data below the required partition value.

```
# to get the first quartile  
prices.quantile(.25)
```

34.5

```
# to get the second quartile  
prices.quantile(.50)
```

56.0

```
# to get the third quartile  
prices.quantile(.75)
```

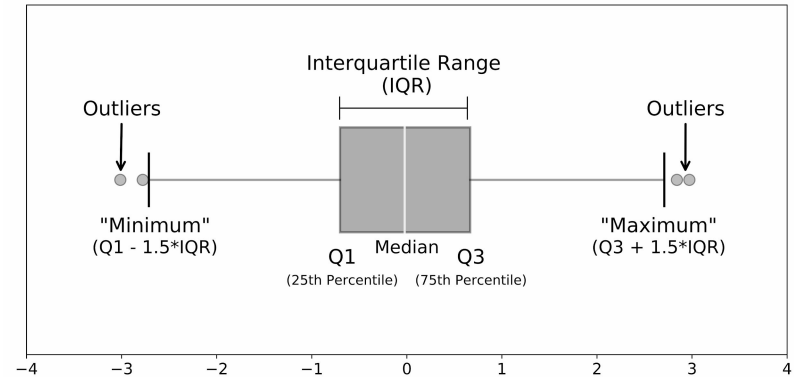
88.0

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Understanding box plot

- A box plot is a way of displaying the distribution of data based on a five number summary (minimum, first quartile (Q1), median (Q2), third quartile (Q3), and maximum)
- The whiskers give an idea of the spread of the data
- The dots outside of the whiskers are the outliers



Interquartile range

- The interquartile range is defined as the difference between the third and the first quartile
- It gives the range in which the middle 50% of the data lies
- The value of IQR is used to remove outliers in the data

$$\text{IQR} = Q_3 - Q_1$$

Deciles

- The values that divide the dataset into ten equal parts are called deciles
- 10% of the data lies below the first decile (D1) and 90% above it
- Similarly, 20% of the data lies below the second decile (D2) and 80% above it and so on
- The fifth decile is the same as the second quartile i.e. $D5 = Q2 = \text{Median}$
- There are in all 9 deciles

Measures of Dispersion

Measures of Dispersion

- The measure of dispersion refers to the variability within the data
- Variability is the measure of how close or far the data lie from the central value
- Reliability of a measure of central tendency is more if dispersion in the data is less
- It includes range, variance, standard deviation, coefficient of variation and interquartile range

Range

- Range is the difference between the largest and the smallest observation
- It is defined as

$$\text{Range} = X_n - X_1$$

where X_n is the largest value and X_1 is the smallest value

Range

Merits :

- It is a basic way to measure the variability

Demerits :

- If either the smallest or the largest value turns out to be an outlier then the range value is unreliable
- The other in-between observations do not affect the range value
- It does not give a clear picture of variability in the data

Obtain the range

```
# Series 1  
year_establishment = pd.Series((1955, 1900, 1980, 1990, 2000))  
Range = year_establishment.max() - year_establishment.min()  
Range
```

100

```
# Series 2  
year_establishment = pd.Series((1920, 1990, 1925, 1930, 2020))  
Range = year_establishment.max() - year_establishment.min()  
Range
```

100

Use the formula for range

Interpretation: The range of both the variables is 100. Thus, range is not a good measure of dispersion.

Variance

Variance is the arithmetic mean of squares of deviations taken from the mean. It is given as

Formula for Variance:

For a population:

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2$$

For a sample:

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

Interpretation: The variance looks at how spread out the observations are from the mean. The higher the variance more the data is spread out.

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Obtain the variance

```
# obtain the variance for the variable 'prices'  
prices.var()
```

```
1054.3290322580644
```

More the data is spread out away from the mean, higher the variance.



Suppose we have a dataset representing the ages of five people: [25, 30, 35, 40, 45]. Find the Variance

The variance of in terms of Population is 50
for the sample is 62.5

It indicates how much the ages vary or dispersed around the mean age of 35.



The unit of measurement of variance

- The unit of measurement of the variance is not the same as that of the original data
- Consider an industry producing screws. The diameter of each screw is noted in millimeters
- The variance in the diameter is measured in squared millimeters
- Thus a usual practice is to take square root of the variance which would have the same unit as the original data
- The value thus obtained is called the standard deviation of the data

Standard deviation

The standard deviation of the variable is the square root of the variance.

Formulas:

1. For a Population:

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2}$$

Where:

- σ = Population standard deviation.
- N = Number of data points in the population.
- x_i = Each individual data point.
- μ = Population mean.

2. For a Sample:

$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$$

Where:

- s = Sample standard deviation.
- n = Number of data points in the sample.
- x_i = Each individual data point.
- $\bar{x}$ = Sample mean.

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Obtain the standard deviation

```
prices.std( )
```

```
32.470433200960905
```

Interpretation: The standard deviation looks at how spread out the observations are from the mean. The higher the standard deviation more the data is spread out.

Coefficient of variation

- The coefficient of variation is a statistical measure of dispersion of data points around the mean

Formula:

$$CV = \left(\frac{\sigma}{\mu} \right) \times 100\%$$

Where:

- σ = Standard deviation of the dataset.
 - μ = Mean of the dataset.
- It is always expressed in percentage and is unit free
 - Generally, used to compare dispersion of two or more groups
 - Measures relative variability and is expressed as a percentage of the mean.
 - According to Karl Pearson, CV is the percentage variation in the mean and S.D. is the total variation in mean

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Obtain the coefficient of variation

```
# importing variation from scipy.stats  
from scipy.stats import variation  
  
scipy.stats.variation(prices)  
  
0.5597598237090757
```

The variation() from the scipy library computes the coefficient of variation.

Note: The name of the function can be quite misleading; do not confuse with 'variance'.



Example Calculation:

Let's say we have two different products, and we want to compare the consistency of their weights based on a sample of measurements.

- Product A: Weights = [10, 12, 14, 13, 11]
- Product B: Weights = [100, 102, 98, 105, 101]
 - Step 1 :
 - Product A Mean = 12
 - Product B Mean = 101.2
 - Step 2:
 - Standard Deviation: $\sigma_A \approx 1.414$
 - Standard Deviation: $\sigma_B \approx 2.315$
 - Step 3 :
 - **Product A** has a CV of approximately **11.78%**.
 - **Product B** has a CV of approximately **2.28%**.

Interpretation:

This means that Product B has less relative variability in its weights compared to Product A, even though the absolute standard deviation of Product B is higher. The CV allows us to make this comparison meaningful, especially when the means of the datasets are different.

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Summary

- Range is not a reliable measure as it do not consider the in-between observations
- The variables with a near-zero standard deviation have less significance in the analysis
- Coefficient of variation (CV) is the unitless quantity
- Box-plot is used to visualize five-number summary (minimum, Q1, Q2, Q3, and maximum)
- IQR can be used to detect and remove the outliers in the data

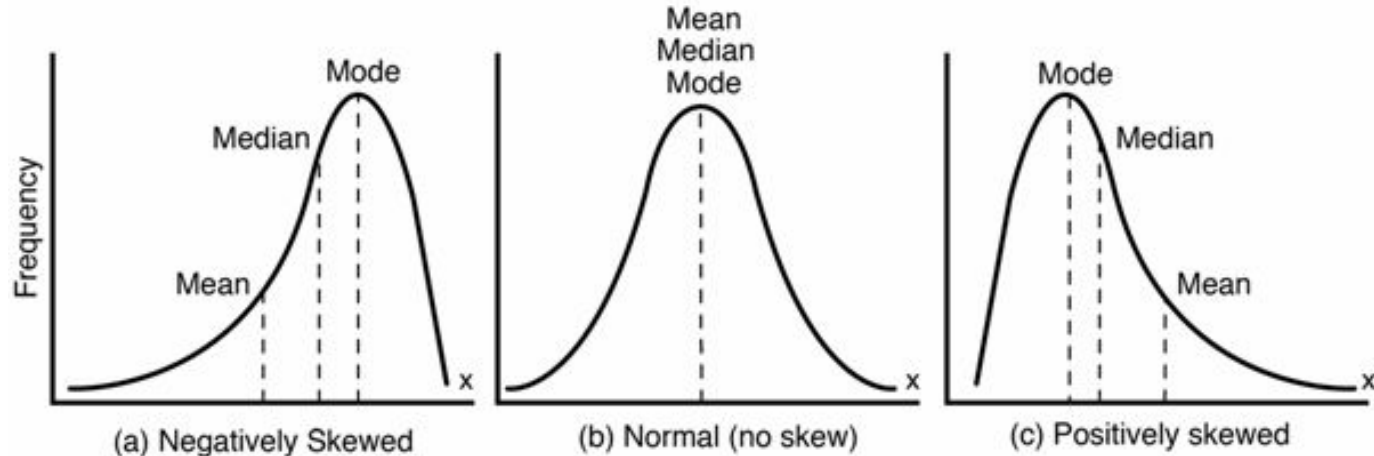
Skewness

Skewness

- Skewness is a lack of symmetry or departure from symmetry
- If the distribution of the data is elongated on either side then the data is said to be skewed
- If the distribution of the data is elongated on the left side then the data is said to be left skewed
- If the distribution of the data is elongated on the right side then the data is said to be right skewed

Skewness

The graph of skewed distribution



mean < median < mode

mean = median = mode

mean > median > mode

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Calculate the skewness

The Bowley's Coefficient of Skewness is defined as

$$S_b = \frac{Q_3 - 2Q_2 + Q_1}{Q_3 - Q_1}$$

where Q_1 , Q_2 and Q_3 are first, second and third quartiles respectively

Value of skewness	Interpretation
$S_b < 0$	The distribution is negatively skewed
$S_b = 0$	The distribution is not skewed (symmetric distribution)
$S_b > 0$	The distribution is positively skewed

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Calculate the skewness

The Bowley's coefficient of skewness:

```
# calculate Bowley's coefficient of skewness
# calculate first quartile
Q1 = prices.quantile(0.25)

# calculate second quartile
Q2 = prices.quantile(0.5)

# calculate third quartile
Q3 = prices.quantile(0.75)

s_b = (Q3 - 2*Q2 + Q1) / (Q3 - Q1)

# Bowley's coefficient of skewness
print(s_b)

0.19626168224299065
```

Interpretation: The coefficient is positive but near zero. Thus, the variable 'prices' is said to be near symmetric.

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Calculate the skewness

The Karl Pearson's Coefficient of Skewness is defined as

$$S_k = \frac{3(\text{mean} - \text{median})}{\sigma}$$

Where σ is the standard deviation

Value of skewness	Interpretation
$S_k < 0$	The distribution is negatively skewed
$S_k = 0$	The distribution is not skewed (symmetric distribution)
$S_k > 0$	The distribution is positively skewed

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Calculate the skewness

The Karl Pearson's coefficient of skewness:

```
# calculate Karl Pearson's coefficient of skewness
# calculate the mean
price_mean = prices.mean()

# calculate median
price_med = prices.median()

# calculate standard deviation
price_sd = prices.std()

s_k = (3*(price_mean - price_med)) / price_sd

# Karl Pearson's coefficient of skewness
print(s_k)

0.09835250325524642
```

Interpretation: The coefficient is positive but near zero. Thus, the variable 'prices' is said to be near symmetric.

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Calculate the skewness

- The adjusted Fisher-Pearson Coefficient of Skewness is defined as

Formula:

For a sample:

$$g_1 = \frac{n}{(n-1)(n-2)} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s} \right)^3$$

- This coefficient of skewness is based on the central moments of the distribution

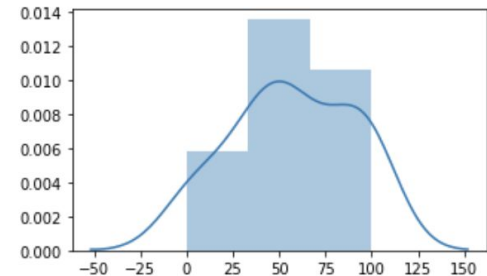
Obtain the skewness

```
# use the 'skew()' to calculate the skewness  
prices.skew()
```

```
-0.20497188990720197
```

Interpretation: The variable prices is slightly negative skewed. It can be said it is close to symmetric.

```
# import the required libraries  
import seaborn as sns  
import matplotlib.pyplot as plt  
  
# set the plot size  
plt.figure(figsize=(5,3))  
  
# plot a distribution plot  
sns.distplot(prices)  
  
# display the plot  
plt.show()
```



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Coefficient of skewness

- The magnitude of the coefficient of skewness explains the extent of skewness in the distribution
- The +/- sign of a coefficient provides the direction of skewness
- The coefficient compares the distribution of the sample to the normal distribution. Thus, the zero value represents the symmetric distribution
- A higher magnitude of the coefficient indicates that the distribution highly differs from the normal distribution

Summary

Bowley's Coefficient of Skewness

- Based on quartiles
- Value lies between -1 to +1
- Can be used to calculate skewness for open-ended distribution
- Preferred if the data has extreme values (outliers)

Karl Pearson's Coefficient of Skewness

- Based on mean, median and standard deviation
- Usually value lies between -3 to +3
- Can not be used for open-ended distribution, as we can not calculate the mean in such case

Fisher - Pearson Coefficient of Skewness

- Based on central moments
- Mostly used measure of skewness by different softwares
- Can not be used for open-ended distribution, as we can not calculate the mean in such case

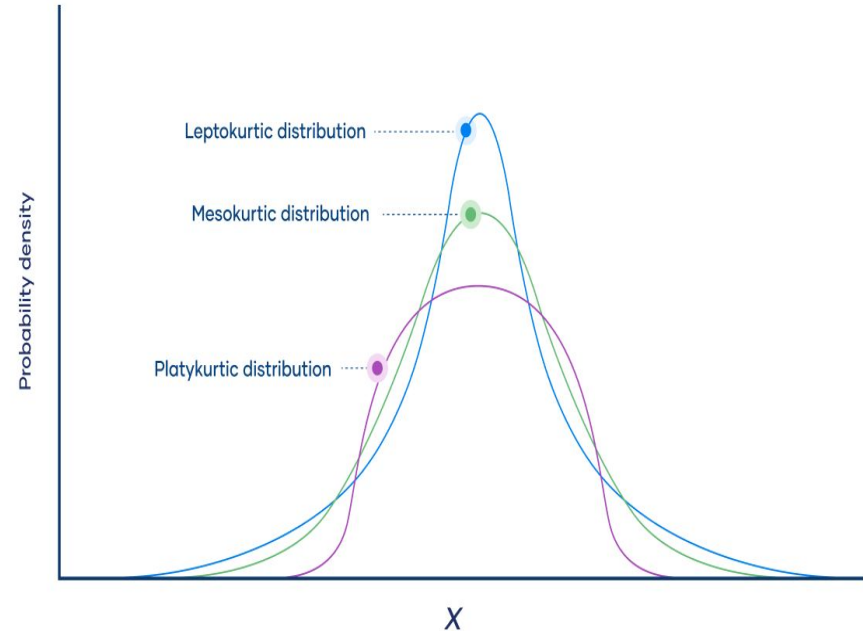
Kurtosis

Kurtosis

- Kurtosis measures the peakedness of the distribution
- In other words, it is a statistical measure that defines how the tails of the distribution differ from the normal distribution
- Kurtosis identifies whether the tails of a given distribution contain extreme values
- A histogram is an effective way to show both the skewness and kurtosis of a data set

Kurtosis

Value	Interpretation
Kurtosis < 0	The distribution is platykurtic
Kurtosis = 0	The distribution is mesokurtic
Kurtosis > 0	The distribution is leptokurtic



Kurtosis

Leptokurtic (Heavy-Tailed Distribution):

- **Description:** A leptokurtic distribution has a kurtosis value greater than 0, which indicates heavier tails and a higher peak compared to a normal distribution. This means that the distribution has more frequent extreme values (outliers) and a sharper peak around the mean.

Mesokurtic (Normal Distribution):

- **Description:** A mesokurtic distribution has a kurtosis value close to 0, which indicates that its tails are similar to those of a normal distribution. The shape of the distribution has a moderate peak and tails that are neither too heavy nor too light.

Platykurtic (Light-Tailed Distribution):

- **Description:** A platykurtic distribution has a kurtosis value less than 0, which indicates lighter tails and a flatter peak compared to a normal distribution. This means that the distribution has fewer extreme values and is more spread out.

Obtain the kurtosis

We use `DataFrame.kurtosis()` to find the kurtosis

Interpretation: The variable 'prices' is platykurtic since $-1.036 < 0$. This means there are more dispersed prices with lighter tails. i.e. less extreme values.

```
prices.kurt()
```

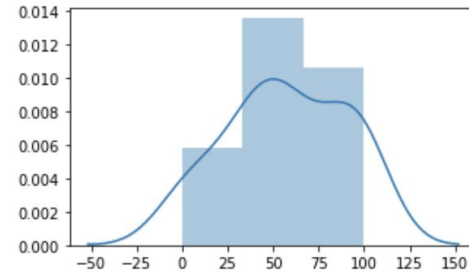
```
-1.0366224980536156
```

```
# import the required libraries
import seaborn as sns
import matplotlib.pyplot as plt

# set the plot size
plt.figure(figsize=(5,3))

# plot a distribution plot
sns.distplot(prices)

# display the plot
plt.show()
```



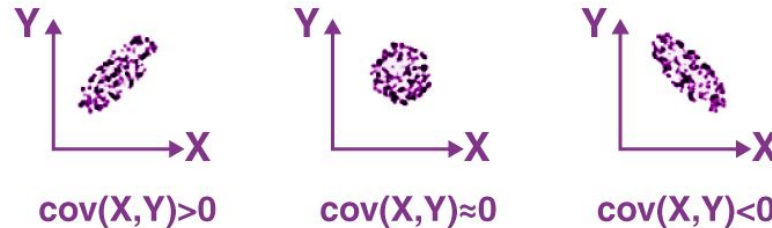
Summary

- A mesokurtic distribution is basically a normal distribution
- A leptokurtic distribution has heavy tails on both the sides that indicates the presence of large outliers. Also the distribution is concentrated towards the mean than the normal distribution
- A platykurtic distribution has flat/lighter tails on both the sides and it indicates that there are small amount of outliers in the distributions. Also, the values are more dispersed and fewer values are close to the mean

Covariance

Covariance

- Covariance is a measure of how two variables change together. It indicates the direction of the linear relationship between the variables, but it doesn't provide the strength or magnitude of the relationship in a standardized way.
- Covariance is similar to variance, but variance explains how a single variable varies, and covariance explains how two variables vary together.
- It cannot tell whether the value indicates a strong relationship or weak relationship, since the covariance can take any value
- Covariance can theoretically range from $-\infty$ to $+\infty$.



Covariance

The covariance for two variables X and Y are given by

$$\text{Population Covariance: Cov}(x, y) = \frac{\sum_{i=1}^N (X_i - \mu_x)(Y_i - \mu_Y)}{N}$$

$$\text{Sample Covariance: Cov}(x, y) = \frac{\sum_{i=1}^N (X_i - \bar{x})(Y_i - \bar{y})}{n - 1}$$

Obtain the covariance

```
# create imports series
imports_raw_material = pd.Series([10,11,14,14,20,22,16,12,15,13])

# create exports series
exports_finished_products = pd.Series([12,14,15,16,21,26,21,15,16,14])

# calculate the covarinace
imports_raw_material.cov(exports_finished_products)

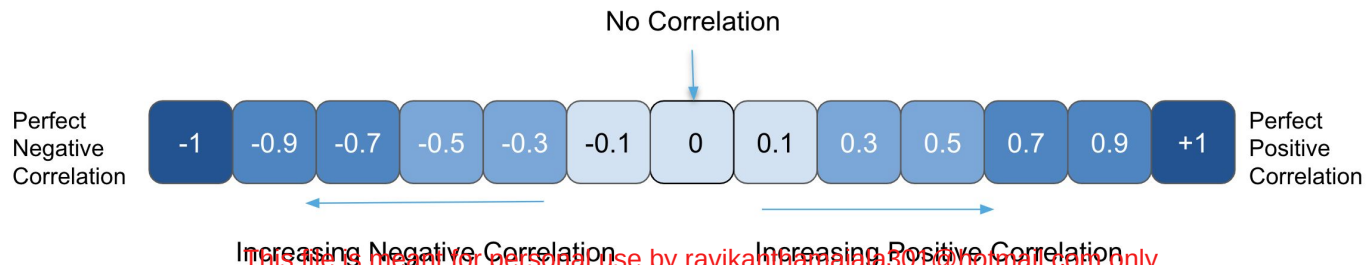
15.444444444444443
```

Interpretation: There is a positive covariance between imports of raw material and export of finished goods

Correlation

Correlation

- Correlation is a statistical measure that describes the strength and direction of the linear relationship between two variables.
- Unlike covariance, which is in units derived from the original variables, correlation is a standardized measure and ranges from -1 to $+1$.
- It indicates the extent to which two variables increase or decrease in parallel
- The value of a correlation coefficient ranges between -1 and 1

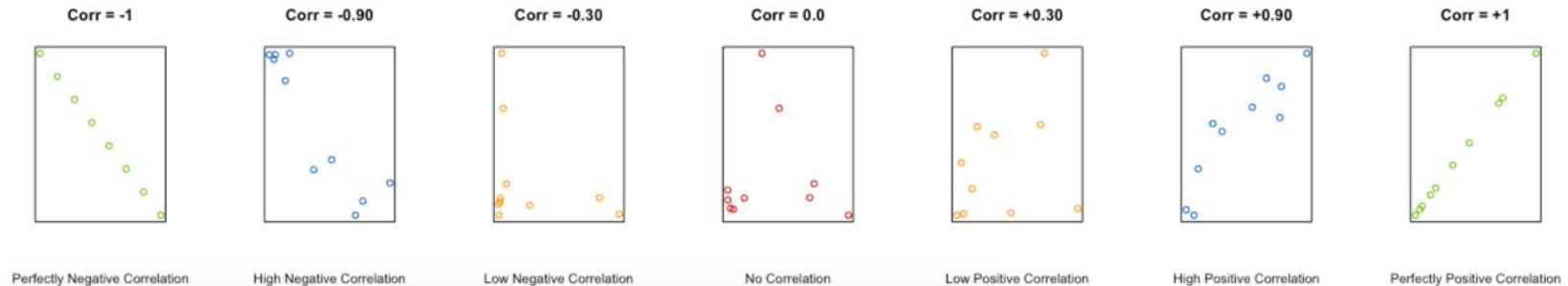


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Correlation

- A positive correlation exists if variable increase/decrease simultaneously
- A negative correlation exists if one variable increases, while the other variable decreases



Obtain the correlation

```
# create imports series
imports_raw_material = pd.Series([10,11,14,14,20,22,16,12,15,13])

# create exports series
exports_finished_products = pd.Series([12,14,15,16,21,26,21,15,16,14])

# calculate the correlation
imports_raw_material.corr(exports_finished_products)

0.9458496510416226
```

Interpretation: There is a high positive correlation between imports of raw material and export of finished goods

Correlation

- Correlation is normalized form of covariance

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_x \sigma_y}$$

where σ_x is the standard deviation of X and σ_y is the standard deviation of Y

- The two terms conceptually they are almost identical

Covariance and correlation

Covariance

- tells whether two variables are related by measuring how the variables change in relation to each other
- Gives the direction of the relationship

Correlation

- Similar to covariance
- Its tells strength and how strong is the relationship

Summary

Summary

Measures of central tendency

A measure of central tendency is a single value that identifies the central position of the data

Measure of Central Tendency	Description	Example
Mean	Ratio of sum of all observations to the number of observations	price = [1, 2, 3, 4, 5] mean = 3
Median	Middle most observation in the data when it is arranged either in ascending or descending order of their values	price = [6, 8, 9, 18, 20] median = 9
Mode	Value which has the highest frequency	price = [6, 18, 9, 18, 20] mode = 18

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Measures of central tendency

Measure of Central Tendency	Description
Partition values	The values that divide the data into desired number of equal parts
Quartiles	The values that divide the data into four equal parts

Measures of dispersion

The measure of dispersion refers to the variability within the data where variability is the measure of how close or far the data lie from the central value

Measures of Dispersion	Description	Example
Range	The difference between the largest and smallest value in the data	price = [1, 2, 3, 4, 5] range = 4
Variance	The arithmetic mean of squares of deviations taken from the mean	price = [6, 18, 9, 18, 20] variance = 31.36
Standard deviation	The square root of variance	price = [6, 18, 9, 18, 20] Std deviation = 5.6

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Measures of dispersion

Measures of Dispersion	Description
Coefficient of variation	A relative measure of deviation, it is the ratio of standard deviation to the mean
Interquartile Range	The difference between the first and the third quartile

Skewness and kurtosis

	Description
Skewness	Lack of symmetry or departure from symmetry
Kurtosis	Measures the peakedness of the distribution. In other words, defines how the tails of the distribution differ from the normal distribution

Covariance and correlation

	Description
Covariance	Tells whether two variable are related or not and gives the direction of the relationship (positive/negative)
Correlation	Tells whether two variable are related and along with the direction of the relationship (positive/negative), it tells how strong the relationship is

Thank you!

Happy Learning :)